

Formal Verification of Lightning Network Multi-Hop HTLC Routing: From Concrete Models to Arbitrary Path Lengths

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ABSTRACT

The Lightning Network (LN) relies on Hashed Time-Locked Contracts (HTLCs) and complex channel state machines to enable scalable, multi-hop blockchain payments. Prior symbolic analyses have targeted either two-party HTLC settings or generic message-forwarding abstractions; the interaction between the channel revocation lifecycle, multi-hop routing, and block-height timing has not been mechanized as a single symbolic model. We present such a model in the Tamarin prover, structured as a modular package of five theories totalling **81 machine-checked lemmas**, covering the two-party channel lifecycle, multi-hop HTLC routing with fees, staggered-CLTV timing, and an N -hop generalisation.

Our central finding is a consequence of that integration. We identify a *linear single-spend discipline*: two independent soundness gaps—HTLC output duplication and revoked-commitment replay—share one root cause, namely modeling one-shot on-chain resources as Tamarin *persistent* facts rather than *linear* ones, and both are repaired by the same linear-token remedy. The second instance is visible only when revocation and routing are modeled together, which narrower prior models structurally could not observe; we propose the resulting invariant as a reusable modeling discipline for symbolic payment-channel analyses. We further prove economic-soundness theorems (end-to-end conservation, strictly positive fees) and expose a grieving DoS; establish *assumption minimality* empirically, showing that removing the intermediary-claim restriction yields a concrete 59-step counterexample; and reproduce the known wormhole (agent-skipping) attack as a fidelity check on the model. Finally, we identify Tamarin’s origin-analysis recursion on self-referential signed messages as a barrier to unbounded verification, and give a proof-layering technique—machine-checked local authentication plus a pen-and-paper refinement theorem—that transfers core routing invariants to paths of arbitrary length.

CCS CONCEPTS

• **Security and privacy** → **Formal methods and theory of security**; *Cryptography*.

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Formal verification, Tamarin prover, Lightning Network, Payment Channels, HTLC, Blockchain, Layer-2 scaling

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1 INTRODUCTION

Blockchain technologies have revolutionized decentralized finance, but their inherent reliance on global consensus creates a fundamental scalability bottleneck. To achieve the transaction throughput required for global adoption, Layer-2 (off-chain) scaling solutions have become essential. The Lightning Network (LN) [16] is the most prominent of these, enabling fast, low-fee payments by moving the bulk of transactions off-chain and settling only disputes on the underlying blockchain [8].

At the heart of the Lightning Network are Payment Channels and Hashed Time-Locked Contracts (HTLCs) [1]. HTLCs enable multi-hop payments across a network of untrusted intermediaries by conditioning the release of funds on the knowledge of a secret preimage, combined with staggered absolute timelocks (CLTV). Alongside this, the channel lifecycle relies on a complex revocation and punishment mechanism to prevent participants from publishing outdated, fraudulent channel states [10].

While the conceptual design of LN multi-hop routing is elegant, the interaction between cryptographic primitives, block-height timing constraints, and off-chain state machines is highly non-trivial. Subtle edge cases—such as early-timeout races where an incoming channel refunds before an outgoing channel is redeemed—can compromise the safety of honest intermediaries. Furthermore, security in these systems is heavily dependent on liveness assumptions: honest nodes must monitor the network and act before their HTLCs expire.

Existing formal treatments have addressed parts of this stack in isolation. Symbolic analyses in Tamarin have targeted two-party HTLC constructions such as cross-chain swaps [2] and contingent payments [3]; generic symbolic frameworks have studied path integrity and agent-skipping in message-forwarding protocols [17]; and model checking with TLA+ has verified balance security for Lightning channels and bounded-length multi-hop payments [7]. What has not been mechanized is the *combination*: the two-party revocation and punishment lifecycle together with multi-hop HTLC routing, per-hop fees, and staggered CLTV timing, in one symbolic model under a full Dolev–Yao adversary. To our knowledge, the

model we present is the first Tamarin formalization to integrate these layers.

That integration is not merely a matter of coverage; it is what makes our central finding observable. We identify a category error—modeling one-shot on-chain resources as *persistent* rather than *linear* Tamarin facts—that manifests twice, once in the HTLC layer and once in the revocation layer. The second instance involves a rule gated by revocation state being fired repeatedly along a routing execution, and is therefore invisible to models that contain only one of the two layers.

To conduct this analysis we use the Tamarin prover [14] under a full Dolev–Yao adversary [6]. We explicitly model integrity and safety; we do not model payment privacy or unlinkability (a confidentiality property requiring onion routing and observational-equivalence proofs), stated here as an explicit exclusion. Comprehensively verifying the Lightning Network introduces severe methodological and tooling hurdles. First, the protocol inherently requires the combination of Dolev–Yao cryptographic reasoning, natural-number induction (for CLTV arithmetic), and unbounded block clocks. As we demonstrate, attempting to verify these aspects in a single monolithic theory causes state-of-the-art provers to exhaust available memory (OOM). Second, while prior analyses often fix the routing path to a specific length, a natural question arises: do these safety properties hold for paths of arbitrary length (N hops)?

In this paper, we present a comprehensive, modular formal verification of Lightning Network multi-hop HTLC routing. Our contributions are as follows:

- (1) A **linear single-spend discipline**: a category error in which one-shot on-chain outputs are modeled as persistent facts, identified in two independent instances (HTLC output duplication and revoked-commitment replay) and repaired uniformly by structural linear-resource guards, replacing global axioms. We propose the resulting invariant—*every one-shot on-chain output is a linear fact*—as a reusable modeling discipline for symbolic payment-channel analyses.
- (2) A modular, fully machine-checked model integrating the two-party channel lifecycle (including revocation and punishment) with multi-hop HTLC routing and fees, verified across **81 lemmas** in five theories. This integration is the precondition for the finding above.
- (3) An empirical assumption-minimality result proving that removing the intermediary-claim restriction yields a concrete 59-step counterexample, together with a two-sided certificate that the corresponding liveness assumption is load-bearing.
- (4) Economic-soundness theorems (end-to-end value conservation, strictly positive per-hop fees) and a reachable grieving DoS, plus a reproduction of the known wormhole (agent-skipping) attack [12, 17] as a fidelity check on the model.
- (5) A map of the prover’s tractability boundaries, identifying the precise bounding factors (the signing/nat/clock OOM triple, signed-message N -hop routing, and reusable channels), and a proof-layering technique—machine-checked local authentication plus a pen-and-paper refinement theorem—for N -hop invariants.

2 RELATED WORK

2.1 Formal Verification of Payment Channels

Early formalizations with idealized channel closures [5, 11, 15] successfully proved payment atomicity and balance conservation but omitted the revocation and punishment mechanisms required to secure real-world implementations.

Prior symbolic analyses have applied Tamarin to HTLC-based cross-chain swaps [2] and zero-knowledge contingent payments over a public ledger [3]. These models target two-party settings and lack multi-hop routing, channel lifecycle revocation, or staggered CLTV arithmetic. Our work explicitly bridges this gap within a unified, automated framework targeting the multi-hop setting.

2.2 Agent-Skipping and Path Integrity

Smith et al. [17] develop a general symbolic framework, in the multiset-rewriting setting, for *agent-skipping* attacks on message-forwarding protocols, introducing path-integrity goals and analyzing several protocols including payment channel networks. The wormhole attack on HTLC routing [12] is an instance of agent-skipping, and their framework establishes it as a symbolic reachability result. Our treatment of the wormhole is therefore *not* a novel attack finding but a fidelity check: we confirm that our integrated model reproduces the known attack, and additionally tie it to the value layer by quantifying the theft to exactly the bypassed routing fees within a model that also carries channel revocation and settlement. The complementary distinction is one of scope: Smith et al. abstract the forwarding protocol to study path integrity generically, whereas we model the concrete Lightning stack—revocation, punishment, fees, CLTV—in order to reason about properties that depend on those layers.

2.3 Model Checking and Computational Treatments

Grundmann and Hartenstein [7] formalize Lightning in TLA+ and model-check balance security, using stepwise refinement and a time abstraction to make single channels and multi-hop payments checkable separately, scaling to payments over up to four hops with two concurrent payments. Their approach is complementary to ours along two axes. Methodologically, model checking explores a bounded state space with an explicit adversary encoding, whereas symbolic verification in Tamarin reasons over unbounded sessions under an equational Dolev–Yao model; consequently their analysis handles concurrency that we do not, while ours handles unbounded message-derivation capabilities and, via the abstraction of Section 6, unbounded path length. Substantively, their refinement establishes that the current Lightning specification satisfies balance security; our contribution is orthogonal, concerning a modeling pitfall in the symbolic encoding of one-shot on-chain outputs.

Kiayias and Litos [9] provide a composable, game-based (UC) security treatment of the LN at the two-party channel layer; it does not machine-check multi-hop HTLC routing. Our symbolic approach achieves a high degree of automation—proving 81 lemmas—while explicitly covering multi-hop routing under a full Dolev–Yao adversary. Prior work frequently treats the liveness requirement of intermediaries as an implicit assumption. The Tamarin analysis of

HTLC swaps by Boyd et al. [2] surfaced a timing-related hidden assumption (stable relative chain growth), which parallels our finding that liveness assumptions are strictly load-bearing; we provide a mechanized, two-sided certificate for this assumption in a model that includes the full channel lifecycle.

2.4 Tamarin in Blockchain Protocols

The Tamarin prover [14] is a standard tool for verifying cryptographic protocols, successfully applied to industrial standards such as TLS 1.3 [4]. In the blockchain domain, Tamarin has been applied to HTLC swaps and contingent payments [2, 3]. Existing Tamarin applications typically verify fixed-round protocols. Applying Tamarin to stateful, long-running protocols with unbounded loops introduces unique challenges. We do not merely restrict the model to avoid OOM; we rigorously diagnose the root causes, pinpointing Tamarin's origin-analysis recursion on self-referential signed terms as the fundamental barrier to verifying generic N -hop routing.

3 PROTOCOL OVERVIEW AND THREAT MODEL

3.1 Payment Channel Lifecycle

A payment channel is established via an on-chain funding transaction. Once opened, parties transact off-chain by exchanging signed commitment transactions. To prevent a malicious party from broadcasting an outdated state, LN employs a revocation mechanism. When transitioning to a new state, parties exchange revocation secrets. If a revoked state is published, the other party can unilaterally punish the cheater and claim the entirety of the channel funds, protected by a relative timelock (CSV). Channels can be closed cooperatively or unilaterally.

3.2 Multi-Hop HTLC Routing

To pay a recipient without a direct channel, LN uses HTLCs to route payments across intermediate nodes. Assume a path $Sender \rightarrow F_1 \rightarrow \dots \rightarrow F_n \rightarrow Receiver$.

- (1) **The Lock Round (Setup):** The Receiver generates a random secret preimage x and sends its hash $y = H(x)$ to the Sender. The Sender adds an HTLC to their channel with F_1 , locking funds conditional on providing x before an absolute deadline. This cascades down, with each forwarder taking a routing fee.
- (2) **The Release Round (Settlement):** The Receiver reveals x to claim the final HTLC. F_n uses x to claim the HTLC from F_{n-1} . This cascades backward to the Sender.

3.3 Threat Model and Security Goals

We adopt the standard Dolev-Yao adversary model [6], representing a network attacker with full control over the communication medium but who cannot break cryptographic primitives. We explicitly model long-term key compromise as a distinct adversarial capability.

The Liveness Assumptions. Security degrades into a liveness requirement. Our model encodes four timing restrictions:

- **T1** (RedeemBeforeTimeout);

- **T2a** (ForwardTimeGap);
- **T2b** (HonestPartiesActBeforeIncomingTimeout);
- **T3** (IntermediaryMustClaim).

As we demonstrate in Section 5, T3 is strictly load-bearing. This mirrors the real-world necessity of LN watchtowers [13]. Our core security goals translate directly to verified lemmas: Payment Atomicity, Intermediary Safety, Routing Causality & Authentication, Timing Safety, and Value Conservation.

4 FORMAL MODELING METHODOLOGY

4.1 The Curse of Monolithism and Modular Decomposition

A naive approach is to construct a single, monolithic theory containing the channel lifecycle, HTLC routing logic, CLTV block-height arithmetic, and value conservation. This is fundamentally intractable. The combination of the signature equational theory, natural-number induction, and an unbounded block clock causes Tamarin's search space to explode. Crucially, this was tested, not assumed—any two of the three ingredients remain tractable; only the full triple OOM-kills the prover.

The model is therefore split so that no single theory carries all three. The architecture is summarized in Table 1.

4.2 Tactic Engineering and Reproduction

All results were produced under a strict test-first discipline: a property is claimed only after Tamarin reports it. Per-lemma invocation is used on the large theory to avoid OOM. The flag `--stop-on-trace=seqdfrs` rescues the heavy existence witnesses but hangs the natural-number arithmetic theories, so it is applied selectively.

5 VERIFICATION OF THE CORE MODEL

Across the five theories detailed in Table 1, we successfully prove 81 lemmas. Table 2 maps the lemma groups in our primary concrete theory.

5.1 The Linear Single-Spend Discipline (Axiom to Structure)

We identified a category error causing two independent soundness gaps: modeling one-shot on-chain resources (HTLC outputs and revoked commitment outputs) as Tamarin **persistent facts** rather than **linear facts**. A persistent fact is never consumed, so a rule gated only by such a fact may fire arbitrarily often; when that fact stands for an output that can be spent exactly once, the encoding silently permits multiple spends.

Instance 1: HTLC Output Duplication. The forwarding rules were gated only by a public HTLC-offer message. A Dolev-Yao replay of that offer could re-fire an honest forward, locking a second HTLC on the same channel output with an adversary-chosen amount. We replaced a global `OneHTLCPerPtr` axiom with a structural guard: each channel open emits one linear `Free(ptr)` token, and every HTLC-add consumes it.

Instance 2: Revoked Commitment Replay. The punishment rules were gated only by a persistent `!RevokedSecret` fact. Because persistent facts are never consumed, the punishment rule could fire arbitrarily many times for the same revoked state, allowing repeated

Table 1: The five-theory modular architecture (81 lemmas total).

Theory File	Aspect Modeled	Lemmas	Builtins
multihop.spthy	Lifecycle, 3-hop routing, value/fees, wormhole	43	Hashing, Signing, Nat
multihop_nhop.spthy	N-hop generalisation (generic linear-fact forward)	25 (+4 comm.)	Hashing, Signing, Nat
Clock.spthy	Block-clock lifecycle & staggered-CLTV claim windows	9	Natural-numbers
cltv_blocks.spthy	Pure CLTV block arithmetic (delta positivity)	3	Natural-numbers
timeout.spthy	Early-timeout race (T2b removed)	1	—

Table 2: Lemma groups in multihop.spthy (43 total).

Group	Class	Representative Lemmas
Invoice / preimage auth	Safety	Invoice_Authenticates_Settlement
Lock causality	Safety	Forward1_Requires_Offer
Release causality	Safety	Fulfill_Requires_Forward2
Atomicity / non-vacuity	Safety/Reach.	Payment_Atomicity_Under_Liveness
Value & fees	Safety	EndToEnd_Value_Conservation
Refund / grieving	Reachable	Honest_Intermediary_Refunds
Wormhole	Attack	Wormhole_Steals_Exactly_The_Fees
Intermediary safety	Safety	Intermediary_Never_Loses_Under_Liveness
Channel lifecycle	Safety	No_Punishment_Without_Cheating

fund extraction from a single cheat. We fixed this by emitting a linear `RevokedCommitmentUnspent` token in the revocation rule, which is consumed upon punishment. Persistent *knowledge* of the revocation secret is retained via `!RevokedSecret`; only the one-shot *spend* is made linear.

We stress that Instance 2 is a consequence of integration rather than of scale. It resides at the junction of the revocation lifecycle and the routing execution: the offending rule is gated by revocation state yet re-fired along a routing trace. A model containing only the two-party channel layer, or only an abstracted forwarding layer, does not exhibit the trace that exposes it. This is the sense in which the breadth of the model is a precondition for the finding rather than an end in itself.

After these changes, “at most one spend per output” is a consequence of linear-fact consumption—one fewer assumption. Adding an input fact restricts a rule’s firings and therefore preserves all-traces safety properties automatically; it can, however, invalidate an exists-trace witness, so the theory was fully re-run rather than argued by monotonicity. All 43 lemmas re-verify after the change (reported: 262 s, Tamarin 1.8).

The reusable invariant. Both instances share one root cause and one remedy, which we state as a modeling discipline: *every one-shot on-chain output must be modeled as a linear fact*. Persistent facts may encode durable *knowledge* (keys, revealed secrets, channel identity), but a *spend opportunity* must be tracked by a linear token consumed on use. This yields a mechanical audit question for symbolic payment-channel models: for each on-chain output that ought to be single-spend, is there a linear fact consumed when it is spent?

5.2 Wormhole Attack: A Fidelity Check

The wormhole attack [12] occurs when two colluding path members relay the preimage out-of-band, skipping an honest intermediary who forwards value but earns no fee. It is an instance of the agent-skipping class formalized symbolically by Smith et al. [17], and we claim no novelty for exhibiting it. We include it as a fidelity check: a model of multi-hop routing that did *not* admit the wormhole would be suspect. In the symbolic model, the Dolev-Yao network is the colluders’ side channel. We machine-check it twice: `Wormhole_Fee_Theft_Reachable` (51 steps) exhibits a trace where the payment settles yet the middle hop is never re-deemed and no intermediary earns a fee, with no key compromise; and `Wormhole_Steals_Exactly_The_Fees` (81 steps) quantifies the theft to precisely the path fees $\%f1 + \%f2$, tying the attack to the value layer of a model that simultaneously carries channel revocation and settlement. Known mitigations (AMHL, PTLCs) are out of scope.

5.3 Economic Soundness and Grieving

Three all-traces theorems: `EndToEnd_Value_Conservation` (the sender locks the receiver’s amount plus a strictly positive total fee; no value created); `Fee_Strictly_Positive_Hop1/2` (every hop’s fee is > 0 —structural, because the natural-number sort has no $\%0$); and, as a reachability result, `Honest_Intermediary_Refunds` (an honest intermediary induced to lock capital then refund on timeout—the standard PCN grieving DoS).

5.4 Timing Safety and Assumption Minimality

`cltv_blocks` derives the CLTV-delta invariant from concrete block numbers; `Clock` adds block-clock lifecycle safety; and `timeout` exhibits the early-timeout race that reappears once T2b is removed.

A natural question is whether the intermediary-claim restriction T3 is redundant given T1 and T2b. We tested it by removing T3 and

re-running the suite: `Intermediary_Never_Loses_Under_Liveness` is falsified, with Tamarin finding a concrete 59-step counterexample. T2b orders the two timeouts but does not force the node to sweep the incoming HTLC once its outgoing one is redeemed; that action is exactly what T3 encodes. We have empirically demonstrated that T3 is **load-bearing** for intermediary safety. We tested the removal of T3 only, and accordingly make no claim regarding the individual minimality of T1, T2a, or T2b.

6 GENERALIZING TO ARBITRARY PATH LENGTH (N -HOPS)

Replacing the hardcoded forward rules with one generic rule was attempted in several encodings. A generic signed-message forward causes Tamarin to OOM even for a 1-hop witness due to self-referential origin-analysis recursion.

To bypass this, we abstract the network layer into an internal Tamarin linear fact—`Route(prev, me, ptr, y)`—handed directly from one hop to the next. Because the HTLC is no longer a self-referential signed term, Tamarin's origin analysis no longer recurses. This yields `multihop_nhop.spthy` (25 active lemmas; 4 `[use_induction]` lemmas are commented out as deliberate tractability boundaries because backward chaining through unbounded `Route` facts causes divergence).

Fees are enforced structurally in the N -hop model as a firing guard (`Eq(%vIn, %vOut + %fee)`), meaning end-to-end conservation over N hops is a sum of N fees—not finitely statable in one Tamarin lemma. We note also that the abstraction internalises the network, and therefore deliberately does *not* exhibit the wormhole: the side channel the attack exploits is absent by construction, which is why that result is established in the concrete model.

Refinement Theorem (Pen-and-Paper). Internalising the network removes Dolev-Yao power, so the abstract model A has strictly fewer behaviours than the concrete model C ; abstract safety therefore does not by itself imply concrete safety. The gap is exactly the adversary's ability to forge or replay a forward, and is closed by a *local* per-hop authentication lemma `AUTH`, machine-checked in C (`Forward1_Requires_Offer`, 266 steps; `Forward2_Requires_Forward1`, 274 steps), which quantifies over a single adjacent pair and never over the path or N . Under `AUTH`, any safety property φ holding in A for all N holds in C for all N , by projection and induction over forwarding steps. The refinement argument itself is **pen-and-paper**; Tamarin cannot quantify over N or relate two theories. This is the honest boundary of our verification: both hypotheses are machine-checked, the composition is not.

7 DISCUSSION: WHAT FAILED AND TRACTABILITY BOUNDARIES

We record negative results explicitly—they are a core methodological contribution.

- **The Three-Ingredient OOM.** Signing + natural-number induction + an unbounded block clock together exhaust the prover; any two are fine. This is the reason for the modular split.
- **Signed-Message N -Hop Routing.** Origin analysis on the self-referential `'htlc'` term does not terminate, even for reachability, even with a fresh ID.

- **A Rejected Restriction.** An initial fix `OneRedeemPerPtr` (“at most one redeem per output”) was tested and rejected: it falsified the honest witness `Multihop_Payment_Possible`, because the two Redeemed events in an honest run are the two endpoints' views of one hop, not two spends. The correct guard was `OneHTLCPerPtr` (later made structural).
- **A Caught False Positive.** An exists-trace probe sharing only the payment hash appeared to show “value creation”. Tightening it to a single linked payment chain falsified it—the apparent attack was invoice-hash reuse across two unrelated offers, not a soundness bug.
- **Reusable Channels (Boundary).** Stage 1 (structural single-HTLC-per-output via the `Free(ptr)` slot) ships and re-verifies 43/43. Stage 2 (return `Free(ptr)` on resolution) is a clean tractability boundary: all-traces safety lemmas verify (~ 45 s each), but every exists-trace witness fails to converge. Stage 1 is therefore the shipped model.

8 CONCLUSION AND FUTURE WORK

We delivered a modular, machine-checked model that integrates the Lightning two-party channel lifecycle with multi-hop HTLC routing and fees (81 lemmas over five theories). That integration surfaced a *linear single-spend discipline*: two independent soundness gaps arising from encoding one-shot on-chain outputs as persistent facts, repaired uniformly by linear tokens, and generalizable into an auditable modeling invariant. We further established the minimality of the T3 timing assumption empirically, proved economic-soundness theorems, exposed a griefing DoS, reproduced the known wormhole attack as a fidelity check, and mapped where the prover stops scaling.

The protocol-level safety results confirm expectations—consistent with the composable cryptographic treatment of Lightning [9] and the model-checking results of Grundmann and Hartenstein [7], which our symbolic model complements at the routing layer. The durable contribution here is methodological—a modeling discipline that catches a class of silent encoding errors, together with an account of what it takes to keep Tamarin tractable on a stateful, time-locked, arithmetic protocol, in a regime where every claim (positive or negative) was executed rather than asserted.

Future work includes testing whether the persistent-versus-linear pitfall is latent in other published symbolic models of payment channels and blockchain protocols, which would establish the discipline as a class-level finding rather than a single-model observation; extending the framework to Point Time-Locked Contracts (PTLCs); and bridging to computational verification. Our value conservation proofs verify strict amount invariants, but do not enforce game-theoretic economic rationality (e.g., defending against timelock bribery), which remains outside symbolic reasoning.

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